```
spizza@ubuntu-server-spitzer:~$ ping google.com
PING google.com (172.217.19.174) 56(84) bytes of data.
64 bytes from zrh04s07-in-f174.1e100.net (172.217.19.174): icmp_seq=1 ttl=113 time=21.3 ms
64 bytes from zrh04s07-in-f174.1e100.net (172.217.19.174): icmp_seq=2 ttl=113 time=32.0 ms
^C
--- google.com ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 21.347/26.655/31.963/5.308 ms
spizza@ubuntu-server-spitzer:~$ ping 192.168.0.119
PING 192.168.0.119 (192.168.0.119) 56(84) bytes of data.
64 bytes from 192.168.0.119: icmp_seq=1 ttl=64 time=0.048 ms
64 bytes from 192.168.0.119: icmp_seq=2 ttl=64 time=0.069 ms
^C
--- 192.168.0.119 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1010ms
rtt min/avg/max/mdev = 0.048/0.058/0.069/0.010 ms
spizza@ubuntu-server-spitzer:~$
```

Well known Ports:0-1023

• Registered Ports:1024-49151

• Private Ports (Dynamically allocated Ports) 49152-65535

```
spizza@ubuntu-server-spitzer:~$ nmap 192.168.0.120
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-03-24 10:52 UTC
Nmap scan report for 192.168.0.120
Host is up (0.0023s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT   STATE SERVICE
80/tcp open  http

Nmap done: 1 IP address (1 host up) scanned in 4.34 seconds
spizza@ubuntu-server-spitzer:~$ nmap scanme.nmap.org
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-03-24 10:52 UTC
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.17s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 995 closed tcp ports (conn-refused)
PORT      STATE    SERVICE
22/tcp    open     ssh
80/tcp    open     http
179/tcp   filtered bgp
9929/tcp  open     nping-echo
31337/tcp open     Elite

Nmap done: 1 IP address (1 host up) scanned in 41.30 seconds
spizza@ubuntu-server-spitzer:~$
```

syn scan: nmap -sS 192.168.0.120

```
spizza@ubuntu-server-spitzer:~$ sudo !!
sudo nmap -sS 192.168.0.120
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-03-24 10:57 UTC
Nmap scan report for 192.168.0.120
Host is up (0.00097s latency).
Not shown: 998 closed tcp ports (reset)
PORT   STATE SERVICE
22/tcp open  ssh
80/tcp open  http
MAC Address: 08:00:27:69:5F:D9 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.36 seconds
```

spizza@ubuntu-server-spitzer:~$ sudo nmap -sS -p 1-65535 192.168.0.0/24

ergebnisse:
port 21 => ftp
port 23 => telnet
port 8000 => alternative http
port 9443 => alternative https

sudo nmap -sn 192.168.0.0/24
26 hosts up, 256 addresses scanned

```
spizza@ubuntu-server-spitzer:~$ nmap -A -n -T4 192.168.0.120
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-03-24 11:07 UTC
Nmap scan report for 192.168.0.120
Host is up (0.00055s latency).
Not shown: 998 closed tcp ports (conn-refused)
PORT   STATE SERVICE VERSION
22/tcp open  ssh     OpenSSH 9.6p1 Ubuntu 3ubuntu13.14 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|   256 56:5a:94:09:97:eb:6e:25:4a:eb:fb:29:03:a5:ff:cb (ECDSA)
|_  256 33:28:26:2f:ea:c8:88:2e:7f:d4:5b:f7:0f:7c:b9:49 (ED25519)
80/tcp open  http    Apache httpd 2.4.58 ((Ubuntu))
|_http-title: Apache2 Ubuntu Default Page: It works
|_http-server-header: Apache/2.4.58 (Ubuntu)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/
Nmap done: 1 IP address (1 host up) scanned in 6.99 seconds
```

nmap -A -n -T4 192.168.0.120

```
spizza@ubuntu-server-spitzer:~$ nmap -A -n -T4 192.168.0.120
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-03-24 11:07 UTC
Nmap scan report for 192.168.0.120
Host is up (0.00055s latency).
Not shown: 998 closed tcp ports (conn-refused)
PORT   STATE SERVICE VERSION
22/tcp open  ssh     OpenSSH 9.6p1 Ubuntu 3ubuntu13.14 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|   256 56:5a:94:09:97:eb:6e:25:4a:eb:fb:29:03:a5:ff:cb (ECDSA)
|_  256 33:28:26:2f:ea:c8:88:2e:7f:d4:5b:f7:0f:7c:b9:49 (ED25519)
80/tcp open  http    Apache httpd 2.4.58 ((Ubuntu))
|_http-title: Apache2 Ubuntu Default Page: It works
|_http-server-header: Apache/2.4.58 (Ubuntu)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/
Nmap done: 1 IP address (1 host up) scanned in 6.99 seconds
```

Zusatzfrage: Wie erkennt nmap den Hersteller der Netzwerkkarte? (Intel)
=> MAC Adresse Hersteller teil

```
spizza@ubuntu-server-spitzer:~$ netstat -tuln | grep LISTEN
tcp        0      0 0.0.0.0:22              0.0.0.0:*               LISTEN
tcp        0      0 127.0.0.53:53           0.0.0.0:*               LISTEN
tcp        0      0 127.0.0.54:53           0.0.0.0:*               LISTEN
tcp6       0      0 :::22                   :::*                    LISTEN
tcp6       0      0 :::80                   :::*                    LISTEN
spizza@ubuntu-server-spitzer:~$
```

Option	Funktion
`-t`	Zeigt TCP-Ports an (Transmission Control Protocol)
`-u`	Zeigt UDP-Ports an (User Datagram Protocol)
`-l`	Zeigt nur die Ports, die im Status `LISTEN` sind (lauschen)
`-n`	Zeigt numerische Adressen und Ports an, keine Namen oder DNS-Auflösungen
`	grep LISTEN`